

Digital Forensics and Incident Response (DFIR) Report

Date of Report: Friday, October 2, 2026, 15:43:29

Case Identifier: evidence01.pcap

Subject: Analysis of Network Traffic (Packets 1–572)

1. Executive Summary

An analysis of the network capture `evidence01.pcap` reveals a period of active network discovery, firewall-intercepted broadcast traffic, and a successful outbound SMTP email transmission from an internal workstation (192.168.1.159) to an external AOL mail server. While the email subject lines appear benign, the high volume of data transferred during the SMTP session warrants further investigation for potential data exfiltration.

2. Key Findings

2.1 Email Communication & Potential Data Exfiltration

A complete SMTP session was established on port 587 between the internal host 192.168.1.159 and the external server 64.12.102.142. * **Sender:** sneakyg33k@aol.com (Identified as “Ann Dercover”) * **Recipient:** sec558@gmail.com * **Subject Lines:** “lunch next week” and “rendezvous” * **Observation:** The session involved a significant volume of DATA fragments (approx. 1460 bytes each). The sheer number of packets required to transmit these messages suggests the presence of large attachments or substantial message bodies. * **Status:** The transaction was successfully completed with a 250 OK response from the server.

2.2 Network Security & Firewall Activity

- **Firewall Interception:** The gateway/kernel actively blocked multiple KERN.WARNING events. The firewall dropped UDP broadcast/multicast traffic (NTP and NetBIOS/Browser Service) originating from 192.168.1.159 and 10.1.1.10 directed at 192.168.1.255.
- **Traffic Anomalies:** The presence of 10.1.1.x traffic on a 192.168.1.x segment was noted, suggesting potential VLAN leakage or multi-homed routing configurations.
- **Network Discovery:** Frequent ARP requests and NetBIOS/Browser Service Protocol (BSP) traffic were observed, indicating active device discovery and resource identification by the workstation `ann-laptop`.

2.3 Host-Level Artifacts

- **System Maintenance:** Syslog entries captured Windows Event Log data from `ann-laptop`.
 - **Service Activity:** Logs indicate the stopping of ESENT (Extensible Storage Engine) database processes related to `wuaueng.dll` and `wuauclt`. This is consistent with standard Windows Update service maintenance.
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3. Asset Inventory

IP Address	Role/Identity	Notes
192.168.1.159	Workstation (<code>ann-laptop</code>)	Primary source of SMTP traffic and NetBIOS broadcasts.
192.168.1.10	Gateway / Local Host	Acting as NTP server; logging firewall blocks.
192.168.1.30	Local Host	Target of ARP and NTP synchronization.
64.12.102.142	External Mail Server	AOL ESMTP server (Destination).
10.1.1.20	External DNS Server	Resolved <code>smtp.aol.com</code> for the client.

4. Technical Observations

- **TCP Performance:** During the heavy data burst in the SMTP session, significant fluctuations in the TCP Window Size were observed (dropping from 64240 to 500). This indicates temporary buffer exhaustion or network congestion during the transfer.
 - **Protocol Usage:** Standard protocols identified include ARP, NTP, NetBIOS, NBNS, BSP, and SMTP.
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5. Recommendations

1. **Content Inspection:** Perform deep packet inspection (DPI) or reconstruct the SMTP payloads to determine the exact nature of the attachments sent in the “rendezvous” and “lunch next week” emails.
2. **Network Configuration Audit:** Investigate the source of `10.1.1.x` traffic to rule out unauthorized cross-VLAN communication or misconfigured routing.

3. **Endpoint Verification:** Correlate the observed Windows Update service stops with host-side logs on **ann-laptop** to ensure no malicious persistence was attempted during system maintenance windows.